

An electron thruster for LEO drag compensation using the ionosphere as the return circuit

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Abstract

Small spacecraft below 600 km lose their orbit to atmospheric drag within months to a few years, and most of them carry no propulsion at all. This paper describes a thruster with no tank, no feed system and no neutralizer. A cathode held a few hundred volts below the spacecraft body emits electrons through a single aperture. The escaping beam produces thrust and leaves the body positively charged, so the body collects electrons from the ionospheric plasma until collection equals emission. The circuit closes through the ionosphere with zero net mass flow. Particle-in-cell simulations (WarpX) of a 10 mm conducting body in a dayside ionospheric plasma measure the complete operating envelope over 100–350 V: 3.4 to 43 nN of thrust from 12 to 278 mW of beam power, with 96–99 % of the beam escaping, a floating potential of 4–48 V, and an energy efficiency of 0.67–0.77. Two measured constants describe every run to about 1 %. Against a year of real 2024 drag from orbit propagation, the device cancels mean and maximum drag at 500–600 km for 7–31 mW; at 400 km it covers the mean demand with a slender body. Because both drag and electron collection scale with area, the feasibility condition is independent of vehicle size, and a 3U CubeSat needs 0.9–3.4 W for the same altitudes. A bench experiment confirms the electrode topology on hardware. Open items are the transverse geomagnetic field, the choice of a cold cathode, and the CubeSat sheath regime.

1 Introduction

CubeSats and smaller spacecraft have made LEO cheap to reach, but at 400–600 km their orbits decay in months to a few years and most of them fly without propulsion. The thrust needed to hold altitude on a spacecraft this small is nanonewtons to a few micronewtons, continuously, from tens of milliwatts to a few watts.

Flight-proven micropropulsion reaches that thrust scale. Colloid thrusters on LISA Pathfinder resolved 0.1 μN [1]; NanoFEEP flew on a 1U CubeSat at 0.2–0.3 W [2]. Every one of these spends a stored propellant through a tank, valves, feed lines or precision emitter heads. That hardware costs mass and money, imposes a total-impulse limit, and is rarely justified on a CubeSat.

This paper takes the opposite trade. Every ion and Hall thruster already carries an electron gun as its neutralizer.

The idea is to run that gun alone, with no ion stage: electrons leave the spacecraft through an aperture and the ionosphere returns the same current to the spacecraft skin. Electrons are light, so the thrust per watt is about 200 times lower than a gridded ion thruster. At nanonewtons that penalty costs milliwatts, which a small spacecraft can afford. What it buys is a thruster made of two electrodes and a high-voltage supply.

The contributions are:

1. a two-constant operating law for the device, measured by PIC simulation over 100–350 V and two body shapes (Sec. 4);
2. a measured collection law for the ambient return current, which fixes the floating potential and the fraction of supply voltage it costs (Sec. 3, 4);
3. a mission assessment against a year of real drag at 400–600 km, and the scaling argument that carries it to CubeSats (Sec. 5).

2 Concept

The device has three parts (Fig. 1):

1. a cathode, held at $-V$ with respect to the spacecraft body, that emits electrons and accelerates them out through an aperture;
2. the spacecraft's own conducting skin, which is the second electrode and collects electrons from the plasma;
3. a high-voltage supply between the two, which moves collected electrons back to the cathode.

Operation is a single equilibrium, not a cycle:

1. the spacecraft floats in the ionospheric plasma;
2. the supply is switched on and the cathode emits;
3. each escaping electron leaves the body one charge more positive;
4. the positive body attracts thermal electrons from the plasma;
5. collection grows until it equals emission. The body settles at a potential φ where the two currents balance.

Two questions decide whether this works. Why does the positive body not recapture its own beam? Because the

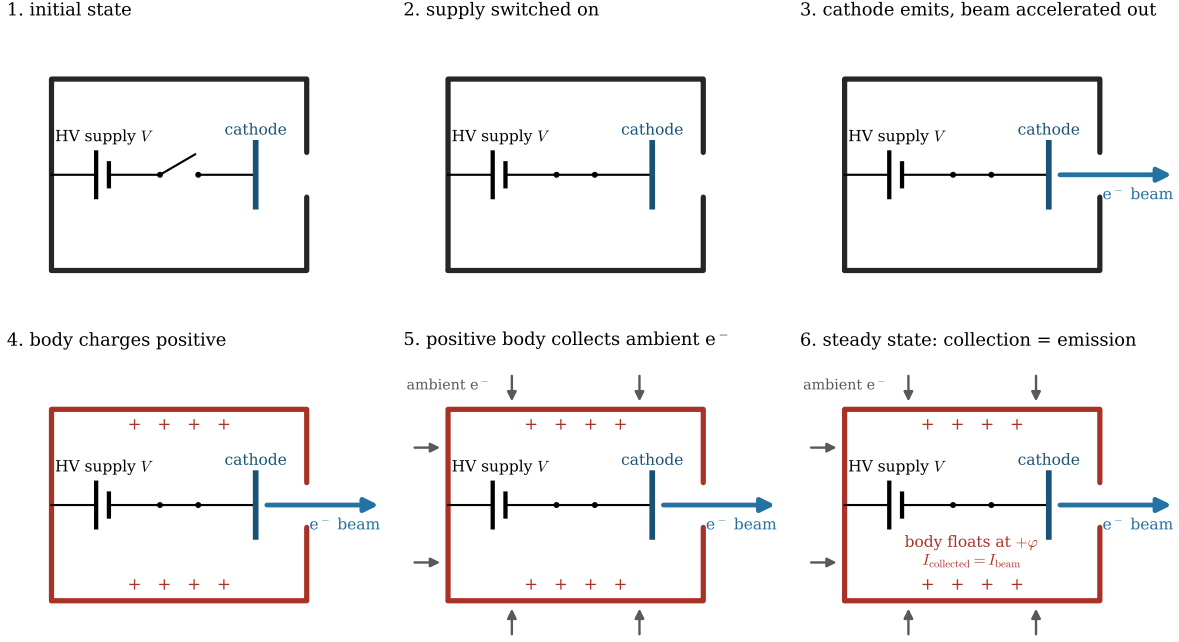


Figure 1: Operating sequence. The supply is switched on, the cathode emits, the escaping beam charges the body positive, and the positive body collects ambient electrons until collection equals emission.

beam electrons carry 100–300 eV while φ is a few volts to a few tens of volts; ambient electrons at 0.1 eV are captured, beam electrons are not. Does the returning current cancel the thrust? Collected electrons arrive with about $e\varphi$ of energy from all directions, against e KE for the beam in one direction. The upper bound on cancellation is $\sqrt{\varphi/\text{KE}}$, about 40 % for straight-astern arrival, and the measured value with isotropic arrival is about 1 % (Sec. 4.5).

3 Theory

3.1 Thrust

The beam is a current I of electrons with kinetic energy KE. Its momentum flux is the thrust:

$$F = \frac{I\sqrt{2m_e \text{KE}}}{e} = 3.37 \frac{\text{nN}}{\text{mA}\sqrt{\text{eV}}} I\sqrt{\text{KE}}. \quad (1)$$

The constant contains only the electron mass and charge. The exhaust velocity is $v_e = \sqrt{2e \text{KE}/m_e}$, of order 10^7 m/s.

3.2 The float tax

The body floats at $+\varphi$ and the cathode at $\varphi - V$, so the accelerating potential is $V - \varphi$. Space charge near the cathode depresses it further. The exhaust energy is

$$\text{KE} = \kappa(V - \varphi), \quad (2)$$

where κ is the gun's energy transmission, set by how hard the gun is loaded relative to its space-charge limit. The ratio φ/V is the fraction of supply voltage spent on collecting the return current.

3.3 Power and efficiency

Beam power is $P = VI$. Jet power over supply power is

$$\eta = \kappa \frac{V - \varphi}{V} f_{\text{esc}}, \quad (3)$$

with f_{esc} the fraction of the beam that clears the body. Eliminating I between Eq. (1) and $P = VI$ at fixed thrust gives $P \propto F\sqrt{V}$: doubling the voltage costs about 40 % more power for the same thrust. Low voltage is cheap, but the current a gun can source is limited by space charge. The planar Child–Langmuir scale for the simulated spot ($r = 0.5$ mm, gap 4.7 mm) is $I_{\text{CL}} \propto V^{3/2}/d^2$, equal to 0.08, 0.24 and 0.43 mA at 100, 200 and 300 V; the real gun emits 1.46 times that. Maximum thrust therefore scales as $I_{\text{CL}}\sqrt{V} \propto V^2$, and at 100 V the spot sources about 0.12 mA whatever the plasma does. The best operating point is the lowest voltage whose emission ceiling still meets the thrust demand. A shorter gap or a wider spot moves the ceiling without changing the voltage. Thrust per watt is $F/P \approx 3.37\sqrt{\kappa}/\sqrt{V}$ in $\mu\text{N}/\text{W}$ with V in volts, about 0.2 $\mu\text{N}/\text{W}$ at 200 V.

3.4 Current collection

The body collects the one-sided thermal electron flux [5]

$$I_{\text{th}} = e A n_e \sqrt{\frac{kT_e}{2\pi m_e}}, \quad (4)$$

enhanced by the positive potential. With $\chi = e\varphi/kT_e$,

$$I = \beta I_{\text{th}} (1 + \chi)^\alpha, \quad (5)$$

where $\alpha = 1$ for a sphere and 0.5 for a long cylinder in orbital-motion-limited theory, and β is a geometry factor.

Setting I equal to the escaping current and inverting gives the float. Its stiffness goes as $1/(n_e\sqrt{T_e})$, so thin night-side plasma makes the body float higher for the same current.

3.5 Scale

Drag is proportional to the ram area; collection and body-mounted power are proportional to the skin area. Thrust demand, beam current and power all scale with ram area, so power per unit ram area depends only on altitude and voltage. Vehicle size drops out and what remains is the shape ratio $A_{\text{skin}}/A_{\text{ram}}$. A 10 mm can and a 3U CubeSat with the same shape ratio have the same feasibility margin.

3.6 Control

I and φ are plain electrical measurements, so Eqs. (1)–(2) let the spacecraft compute its own thrust in flight. Density, temperature and day/night all collapse into where the body floats, so a two-line servo on the measured float is the whole control law. A microcontroller is sufficient.

4 Simulations

4.1 Setup

All runs use WarpX [3] in RZ electrostatic mode with the parameters of Table 1. The body is a closed can with a cathode disk inside and an aperture in the lid, modelled as an embedded conductor (Fig. 2). Its potential is not prescribed. Each step the code counts the charge that left through the aperture and the charge that landed on the skin, and moves the body potential by the net charge over the body’s self-capacitance, which is measured once from Gauss’s law on a vacuum Laplace solve. The float is therefore an output, and ladder stage 7 checks the method on a bare body against the theoretical floating potential. Ions are repelled by the positive body, so the balance is between the escaping beam and collected electrons only; the reduced ion mass affects how fast the sheath settles, not where it settles. The ambient plasma is a dayside 400 km row kept constant by a particle reservoir. Every run freezes its configuration and passes fixed acceptance gates on escape fraction, floating potential, current balance and momentum before it is used.

4.2 Validation ladder

Nine stages build from a vacuum electron gun to the full device, each checked against theory or a known result (Table 2). Stages 1–3 test the gun alone in vacuum: energy conservation across the gap, transmission through a holed anode as space charge and hole size change, and independence from voltage. Stages 4–7 test collection alone on a conducting body in the plasma: thermal flux, enhancement at two bias potentials, and the free float. Stages 8–9 join the two. All nine pass. The transmissions in stages

Table 1: PIC configuration of the 200 V anchor. Characterization runs change one entry at a time (Table 4).

	parameter	value
geometry	body outer diameter $\times$ length	10×5.5 mm
	wall thickness	0.4 mm
	exhaust aperture diameter	4 mm
	cathode disk diameter	3 mm
	emission spot diameter	1 mm
	cathode–lid gap	4.7 mm
electrical	supply voltage V	200 V
	beam current I (prescribed)	0.342 mA
	gun loading I/I_{CL}	1.46
	body self-capacitance	0.5–1 pF
plasma	n_e	$1.63 \times 10^{12} \text{ m}^{-3}$
	kT_e	114 meV
	kT_i	81 meV
	ion mass	$400 m_e$
numerics	cell size $\Delta r = \Delta z$	0.15 mm
	domain r_{max}	30 mm
	particles per cell per species	16
	run length	800 ns (29 RC times)
	supply / gun switch-on	100 / 150 ns

Table 2: Validation ladder.

#	stage	result
1	negative cathode, vacuum	35 μV error on 100 V
2	holed anode	$\kappa = 0.97, 0.90, 1.00$ as predicted
3	voltage bracket	transmission independent of V
4	thermal collection	within 1 % of Eq. (4)
5	biased body, +3 V	0.85 of OML ceiling
6	biased body, +10 V	0.81 of ceiling, sheath 4.1→6.9 mm
7	floating body	−0.251 V, inside theory bracket
8	two-node Laplace	exact, 0.0 V violation
9	full device, 200 V	$\varphi = 17.0 \text{ V}$, $f_{\text{esc}} = 98.4\%$, $F = 13.65 \text{ nN}$

2–3 are for lightly loaded vacuum guns; the device gun runs at 1.46 times its space-charge scale, which is where the lower $\kappa = 0.81$ of Sec. 4.3 comes from.

4.3 Voltage envelope

Stage 9 is the anchor. Three further runs at 100, 300 and 350 V hold the same gun loading ($I/I_{\text{CL}} = 1.46$) and the same plasma (Table 3, Fig. 3). Thrust predictions were recorded before each run and landed within 1 %.

Two constants fit all four points:

$$F [\text{nN}] = 3.27 I [\text{mA}] \sqrt{\text{KE} [\text{eV}]}, \quad (6)$$

$$\text{KE} = 0.81 (V - \varphi). \quad (7)$$

The thrust slope is 97 % of Eq. (1); the deficit is plume divergence. $\kappa = 0.81$ is the space-charge depression of a single-aperture gun at this loading. Thrust per watt

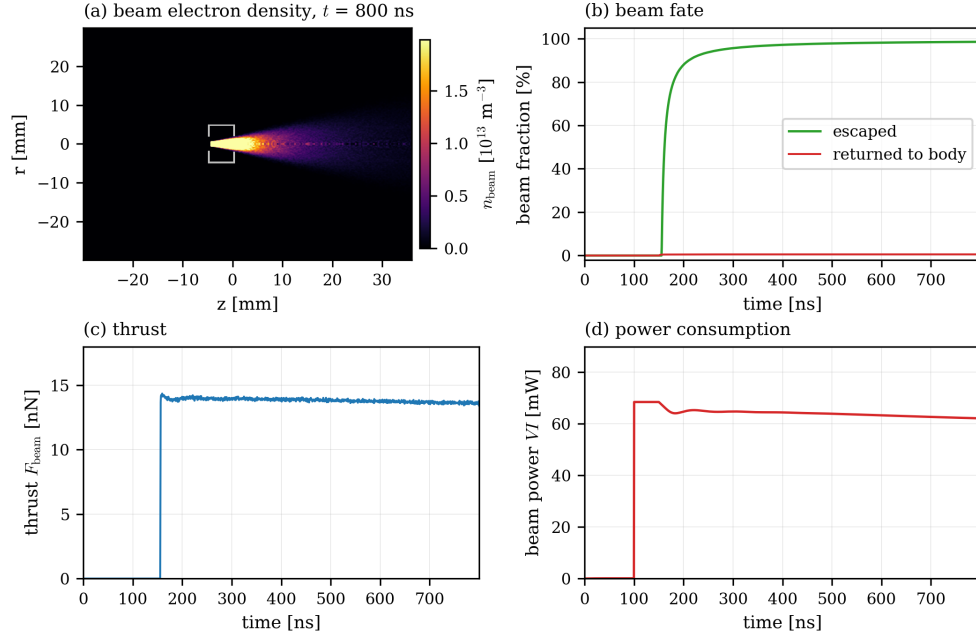


Figure 2: The 200 V anchor run. (a) Beam electron density at the end of the run; the grey outline is the can. (b) Fraction of emitted beam that has escaped the domain or returned to the body. (c) Beam thrust. (d) Beam power. The gun switches on at 150 ns; thrust, power and escape are all steady well before 800 ns.

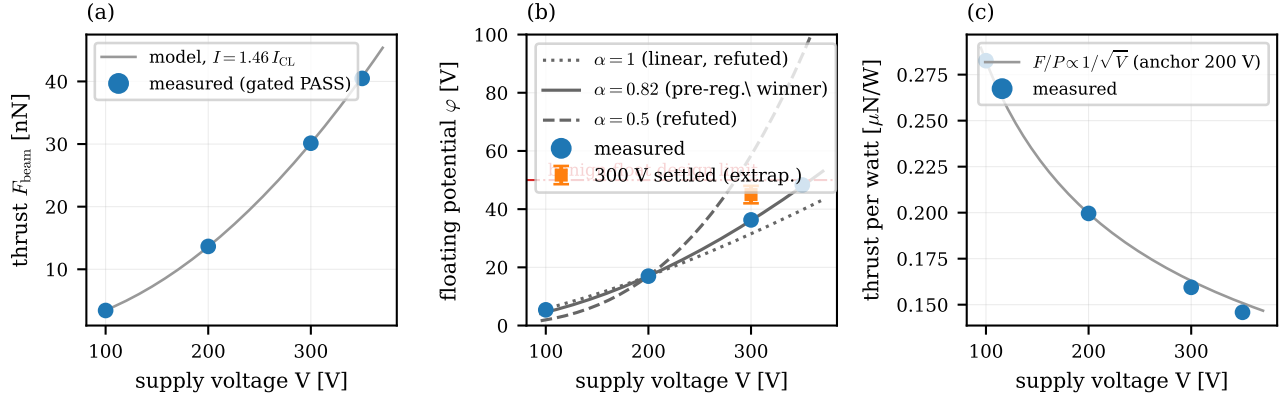


Figure 3: Left: thrust along the envelope. Centre: floating potential against the three candidate collection exponents, anchored at 200 V; the 300 V point selects $\alpha \approx 0.82$ –0.89. Right: thrust per watt against the $1/\sqrt{V}$ law.

Table 3: Measured operating envelope of the 10 mm body. Grid-resolution uncertainty: $F \pm 4\%$, $\text{KE} \pm 7\%$, $\varphi \pm 2\%$.

V V	I mA	φ V	F nN	P mW	f_{esc} %	η
100	0.121	5.4	3.42	12.1	96.1	0.67
200	0.342	17.0	13.65	68.4	98.4	0.73
300	0.630	36.3	30.13	189	99.0	0.74
350	0.793	48.3	40.48	278	99.1	0.74

follows $1/\sqrt{V}$ within 2.5 % over 100–300 V, and the 350 V point sits 3 % below it because the float tax has grown to 14 %. The energy efficiency stays near 0.73 because almost every electron leaves at nearly full energy.

The 300 V run also selects the collection exponent. Three values of α in Eq. (5) agree at 200 V and predict 31, 36 and 90 V for the 300 V float; the measurement is 36.3 V. A fit across the envelope gives $\alpha = 0.82$ –0.89, between the

sphere and cylinder limits. The 350 V float was predicted from that fit at 47 V and measured at 48.3 V. One caveat applies to both high-voltage points: the sheath settles on the ion timescale, and their floats were still rising slowly at run end. The quoted values are tail averages; extrapolating the decaying slope puts the settled 300 V float at 42–48 V and the settled 350 V float just past the 50 V design limit. The compact body at 350 V sits on that limit, which is why the slender body matters at 400 km.

4.4 Characterization

Eight further runs each change one axis of the 200 V anchor (Table 4). All pass their gates.

Shape. Stretching the can to $L/r = 6$ triples the skin at the same ram area. The float drops from 17.0 to 4.4 V and

Table 4: Characterization runs. Each changes one axis of the 200 V anchor except the last, which combines two.

axis	φ [V]	F [nN]	f_{esc} [%]	KE [eV]
anchor (200 V)	17.0	13.65	98.4	147.5
100 V	5.4	3.42	96.1	77.2
300 V	36.3	30.13	99.0	210.1
350 V	48.3	40.48	99.1	239.0
slender body ($L/r = 6$)	4.4	14.22	98.4	159.7
density $n_e/3$	42.5	12.39	99.1	122.0
$B_z = 30 \mu\text{T}$	17.2	13.64	98.4	147.3
$B_z = 300 \mu\text{T}$	49.6	12.06	98.3	115.9
350 V + slender	14.0	43.33	99.1	272.7

the thrust rises 4 %, because a lower float returns voltage to the beam. Before the run, two hypotheses were recorded: area-only scaling predicted 4–5 V, and the long-cylinder limit predicted tens of volts. The measurement confirms the first.

Density. At one third density the float settles at 42.5 V after 2.4 μs , the only fully settled float in the campaign. Every fixed-exponent prediction from Eq. (5) was higher (53–68 V), so the fitted law is conservative in thin plasma.

Magnetic field. An axial field at flight strength (30 μT) changes nothing: float, thrust and escape are within 0.2 V, 0.1 % and 0.06 pp of the anchor. At ten times flight strength, collection is taxed and the float rises 33 V; the thrust law still reproduces the result to within 1 % once the measured float is inserted.

Composition. The slender body at 350 V, the drive the 400 km orbit needs, was predicted from the voltage and shape laws together at 11–17 V and 42–43 nN. It measured 14.0 V and 43.33 nN. The two laws compose, which turns two one-dimensional curves into an operating map over 100–350 V and $L/r = 0.6$ –6.

4.5 Momentum ledger

Each run separates the beam reaction F_{beam} from the net momentum F_{net} deposited by everything that lands on the body: collected electrons, ions, and the returning beam fraction. At 300 V, $|F_{\text{net}}|/F_{\text{beam}} = 1.0$ %; at 100 V, where the beam is slowest, 8.8 %. In a predecessor implementation (14 mm body, 800 V) the full control-volume ledger closed at $F = F_{\text{beam}} - 2.3$ %, and a null run with the body pinned above the beam energy (zero escape) gave zero closed thrust, as momentum conservation requires. The return current does not cancel the thrust.

4.6 Numerical convergence

Doubling the particle count changes every reported quantity by less than 0.05 %. Refining the grid from 0.15 to 0.10 mm raises thrust by 4.0 % and exhaust energy by 7.4 % and lowers the float by 1.8 %. Grid resolution is the lead-

Table 5: Mission cases: 2024 drag on the 10 mm body and the IRI-2020 plasma along the orbit (5th–95th percentile of 5-min samples over the year).

altitude	pose	drag [nN]		n_e [10^{11} m^{-3}]	kT_e [meV]
		mean	max		
400 km	axial	32.9	92.4	2.7–21	87–193
400 km	lateral	21.6	60.7	2.7–21	87–193
500 km	axial	7.6	28.4	1.4–16	91–221
550 km	axial	3.8	16.3	1.0–11	93–233
600 km	axial	2.0	9.6	0.7–7.1	97–244

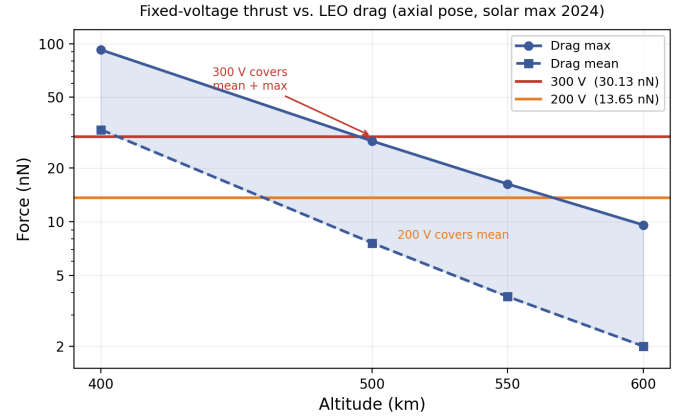


Figure 4: Measured thrust at 200 and 300 V against mean and maximum 2024 drag on the 10 mm body, by altitude.

ing numerical uncertainty; it is carried as the error band in Table 3, and its sign favours the device.

5 Mission assessment

5.1 Method

Drag demand comes from TudatPy [4] propagation over 2024 (solar-cycle maximum) with NRLMSISE-00 density driven by measured F10.7 and A_p , and IRI-2020 plasma along the orbit. The test body is the simulated 10 mm $\times$ 5 mm cylinder at 100 g with $C_d = 2.2$, on a circular near-equatorial orbit ($i = 0.5^\circ$) at 400 km (axial and lateral pose), 500, 550 and 600 km. Drag is cancelled exactly, so the drag at each timestep is the thrust demand. Table 5 summarizes the demand and the plasma each case sees. The PIC plasma row of Table 1 sits near the dayside top of the 400 km range; night-side densities are 5–40 times lower, which is where collection is weakest.

5.2 Coverage

At 500–600 km the thruster covers mean and maximum drag at a fixed 300 V, and in voltage-following mode it costs 7–31 mW (Fig. 4, Tables 6–7). At 400 km the picture is mixed. The compact body delivers 40.5 nN at 350 V against a 32.9 nN mean, but floats on the 50 V charging limit. The slender body delivers 43.3 nN at a 14 V float, covering the mean at about 76 % duty with 3.6 times margin on the limit. The drag maxima (92 nN) exceed any sin-

Table 6: Fixed-voltage coverage of 2024 drag on the 10 mm body.

altitude	mean nN	max nN	200 V 13.65 nN	300 V 30.13 nN
400 km axial	32.9	92.4	–	–
400 km lateral	21.6	60.7	–	mean
500 km	7.6	28.4	mean	mean, max
550 km	3.8	16.3	mean	mean, max
600 km	2.0	9.6	mean, max	mean, max

Table 7: Voltage-following mode: the lowest voltage that cancels each timestep’s drag, and the resulting mean power.

altitude	$V_{\min}$	mean P	in envelope
600 km	100 V	6.8 mW	yes
550 km	104 V	13.4 mW	yes
500 km	146 V	31.3 mW	yes
400 km lateral	247 V	116 mW	yes
400 km axial	304 V	196 mW	yes (350 V)

gle operating point, and the year-round duty for the compact shape is 113 % because thin night-side plasma caps collection. 400 km is a research target, not a closed case. Where full station-keeping does not close, partial compensation still extends mission life.

5.3 Power

Power is the thruster’s only interface to the spacecraft: tens of milliwatts at 500–600 km, 100–200 mW at 400 km. The 10 mm body harvests roughly 10–30 mW from body-mounted cells. A 1U CubeSat has 1–2 W and a 3U 5–10 W. Supply and demand both grow with area, so the ratio holds with size.

5.4 Scaling to CubeSats

By the argument of Sec. 3, a 3U CubeSat flying end-on has the same shape ratio as the slender can and the same feasibility margin (Table 8). In absolute terms it needs 8.8 mA and 0.9 W at 600 km, 1.7 W at 550 km and 3.4 W at 500 km. Larger bodies also need less collection enhancement over the thermal flux (4 times for a 3U at 600 km against 15 times for the 10 mm body), so they rest on less extrapolation, not more. These numbers are estimates from the fitted collection law, and CubeSat radii sit at 25–60 Debye lengths where that law has not been measured (Sec. 8).

6 Comparison with micropropulsion

Table 9 makes three points. The energy efficiency of this device, 0.67–0.77 of supply power in the jet, is higher than any value found for electrospray, FEEP or Hall thrusters, because the exhaust is electrons and nearly all of them leave at nearly full energy. Its thrust per watt is about 200 times below a gridded ion thruster, which is Eq. (1)

Table 8: Feasibility margin (example body-mounted solar supply over demand at 100 V) by shape and altitude. Size cancels; only the skin/ram ratio (in parentheses) matters. Estimates.

body	400	500	550	600 km
power per m ² of ram	1451	335	170	88 W
10 mm can, squat (4)	0.1	0.4	0.8	1.6
1U, face-on (6)	0.1	0.6	1.2	2.3
10 mm can, slender (14)	0.3	1.4	2.8	5.4
3U, end-on (14)	0.3	1.4	2.8	5.4
6U, end-on (12)	0.3	1.2	2.4	4.6

and cannot be engineered away. So the device owns the nanonewton regime: at micronewtons the power penalty reaches watts and electrospray wins; at millinewtons ion and Hall win. Within its regime the difference from every flown system is the hardware column: no tank, no feed system, no neutralizer, no total-impulse limit.

Related concepts share the return-circuit idea. Bare electrodynamic tethers [5, 6] and Janhunen’s electric sail and plasma brake [7, 8] all use continuous electron emission with the ambient plasma closing the circuit, but produce thrust by Lorentz force or ion deflection on kilometre-scale tethers. Atmosphere-breathing electric propulsion [9] collects ambient gas but keeps an ionization stage, an acceleration stage and a neutralizer. A recent review of propellantless propulsion [10] does not list a standalone ionospheric electron-exchange thruster.

7 Bench experiment

The electrode topology was tested on a table. A W5W automotive bulb filament with the glass removed, held at -50 and -62 V on its two terminals, faces a grounded 4×4 cm copper plate in a glass chamber at 4–5 Pa. The ammeter between plate and earth reads a steady 87 mA once the filament is hot: a single net-negative electrode emits electrons and accelerates them across a gap onto a grounded collector. At this pressure the gap is collisional, so the current includes gas amplification and the result is qualitative. It is the hardware counterpart of ladder stage 1; controls (bias off, filament cold) and an I–V sweep are the next step.

8 Limitations

Ordered by how much each could change the answer.

1. **Transverse geomagnetic field.** In LEO the field is roughly perpendicular to the thrust axis. The beam gyroradius (1.4 m) is 50 times the simulation domain, so the fate of the exhaust momentum once the beam gyrates is unmeasured and could move the thrust either way.
2. **No flight cathode.** Beam current is prescribed. The bench emitter is thermionic and its heater uses ten times the thruster’s power; only a cold (field-emission)

Table 9: Position against flight micropropulsion. †: derived from other reported figures.

system	example	thrust	power	I_{sp} [s]	energy efficiency	hardware
this work	electron exchange	3–43 nN	12–278 mW	10^6	0.67–0.77; $F/P =$ 0.15–0.28 $\mu\text{N}/\text{W}$	HV supply, cathode, aperture tanks, valves, nozzles
cold gas	GomSpace	0.1–10 mN	<2 W	60–110	–	
PPT	NanoProp [11] PPTCUP [12]	40 μN	2 W	655	<0.2	capacitor bank, electrodes
electrospray	Busek CMNT [1]	20 μN	16.5 W	225	0.28–0.45†	tanks, PPU, neutralizer
FEEP	NanoFEEP [2]	0–8 μN	0.2–0.3 W	6000	–	liquid-metal needles, heater
FEEP (1U)	Enpulsion IFM Nano [13]	10–500 μN	8–40 W	2000–5000	0.35†	emitter, 2 neutralizers, PPU
RF ion	Busek BIT-3 [14]	0.7–1.2 mN	56–80 W	1400–2640	–	grids, RF cathode, iodine feed, gimbal
Hall	Busek BHT-200 [15]	13 mN	200–250 W	1390	0.44†	channel, hollow cathode, xenon feed

cathode closes the budget, and its cost, lifetime and atomic-oxygen tolerance are open.

3. **One plasma row.** Every run except the $n_e/3$ case uses the same dayside density; below $n_e/3$ the collection law is extrapolation with a measured conservative bias.
4. **CubeSat sheath regime.** Runs sit at $r/\lambda_D \approx 2.5$; CubeSats are 25–60 λ_D across, where orbital-motion-limited collection no longer applies. A single large-body run would settle this.
5. **Reduced ion mass, stationary plasma, held attitude.** Ions are 400 m_e , no ram flow is modelled, and the mission cases assume a fixed pose without an attitude-control model.

All powers are beam power VI ; converter and cathode overheads belong to the flight design.

9 Conclusion

A cathode, a high-voltage supply and the spacecraft’s own skin make a thruster that needs no propellant. PIC simulations measure its operating envelope at 3–43 nN from 12–278 mW, with two constants describing every run, a floating potential fixed by a measured collection law, and a return current that costs about 1 % of the thrust. Against a year of real drag the device holds 500–600 km for tens of milliwatts and covers the 400 km mean demand with a slender body. The feasibility condition depends on shape and altitude but not on size, so a 3U CubeSat inherits it at 0.9–3.4 W. The device is cheap to test: it needs no hardware beyond a supply and an electron gun of the kind that already flies as a neutralizer, and it can be retrofitted to an existing ion or Hall thruster after its propellant is spent. The open questions, the transverse field and the flight cathode, are the next work.

Data. All simulation inputs, run manifests, gate policies, the orbit propagations and the model are committed in the project repository.

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